Primitivity of the $SU(2n)_{2n}$ superconducting series

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Abstract

Gao et al. propose a family of charge-$2ne$ superconductors obtained from $SU(2n)_{2n}$ by identifying its generating fermionic simple current with the physical electron. Their proposed primitivity means that no nonzero flux sector separates into an invertible-superconductor vortex factor and the dynamical anyon sector. We give a uniform argument for every $n > 1$. The full neutral rank exceeds every nonzero-sector rank by at least n. An explicit electron-paired object in each nonzero sector then excludes the remaining Ising-type factor. The rank inequality follows from established cyclic stabilizer resolution and a known roots-of-unity count with positive binomial weights. The construction, definition and counting ingredients are prior work; the claimed advance is their uniform primitivity application with the electron-pair obstruction. Finite computations check the formulas but are not used to establish the universal statement.

1. Statement and conventions

Set $N = 2n \geq 4$, and let $\mathcal{D} = SU(N)_N$. Write J for the generating simple current. We consider precisely the charge-one construction of [1, Sections V.1 and VII.4]: stack the atomic insulator and condense the composite of J with its charge-one local fermion. Write $\mathcal{C}_q$ for the resulting flux sector, $q \in \mathbb{Z}/N$, with physical electron f retained. Put $R_q = |\mathcal{C}_q|$. The set of dynamical anyon representatives has size $R_0/2$. Indeed, transparency and the ribbon identity give $\theta_{fX} = -\theta_X$, excluding $fX \cong X$ for a neutral simple object. Thus electron fusion acts freely on that sector.

Theorem. For every even $N \geq 4$, $R_0 - R_q \geq N/2$ for every $q \neq 0$. Furthermore, every such sector contains two distinct objects interchanged by electron fusion. Consequently none admits an electron-compatible factorization into an invertible-superconductor vortex sector and the dynamical anyon sector. The family is primitive in the sense of [1, Section IV.5].

The term factorization uses the physical electron and object structure of [1], not an arbitrary numerical factorization of ranks. An invertible vortex sector has either two Abelian objects

exchanged by f, or one Ising-type object fixed by f. Allowing both possibilities in every sector only strengthens the exclusion. This includes the Abelian-dressed factors of [1, Sections IV.1–IV.5]: tensoring by an invertible object preserves rank and the electron action. Electron compatibility preserves the specified physical electron, not an arbitrary order-two object. No claim about a microscopic realization or a new family of phases is made.

2. Resolved objects and cyclic stabilizers

Integrable affine labels are

$$X_N = \{(a_0, \dots, a_{N-1}) \in \mathbb{Z}_{\geq 0}^N : \sum_i a_i = N\}.$$

The action of J is cyclic rotation T; the sector is the charge $Q(a) = \sum_i i a_i \pmod N$. Rotation preserves this charge since the total label sum is N. Reversing the monodromy convention replaces q by $-q$ and has no effect on the result.

Let $H = \langle J^2 \rangle$, of order $N/2$, and $A = \bigoplus_{g \in H} g$. The simple-current twist is $\theta_{J^r} = \exp(\pi i r(N-r)) = (-1)^r$. Thus H is bosonic and mutually local, and its regular algebra is condensable. We retain all A-modules, including nonlocal modules representing vortex defects.

Lemma 1 (induction bridge). The resolved objects in the charge-one construction are represented by all A-modules; the retained electron acts as the image of J.

Proof. In $\mathcal{E} = \mathcal{D} \boxtimes \mathrm{sVec}$, let f_{loc} be the external electron and set

$$B = \bigoplus_{r \in \mathbb{Z}/N} J^r \boxtimes f_{\mathrm{loc}}^r.$$

Its summands have twist one and form a bosonic symmetric pointed subcategory, with regular commutative separable algebra B and even subalgebra $A \boxtimes 1$. The action on parent pairs is $(a, \epsilon) \mapsto (T^r a, \epsilon + r \bmod 2)$. Every orbit has parity-zero representatives, related precisely by even rotations; their stabilizers are the H-stabilizers. Resolution is preserved because Frobenius reciprocity identifies

$$\begin{aligned} \mathrm{Hom}_B(\mathrm{Ind}_B(a,1), \mathrm{Ind}_B(b,1)) &\cong \bigoplus_{r \text{ even}} \mathrm{Hom}_{\mathcal{D}}(a, J^r b) \\ &\cong \mathrm{Hom}_A(\mathrm{Ind}_A\, a, \mathrm{Ind}_A\, b). \end{aligned}$$

Odd terms vanish since $\mathrm{Hom}_{\mathrm{sVec}}(1, f_{\mathrm{loc}}) = 0$, and composition agrees through the even subalgebra. Every module is a summand of an induced module; odd-parity induction is rewritten using Ja with parity zero. Passing to these summands identifies resolved objects. Electron fusion changes parity, hence gives J or its inverse on parity-zero representatives; their images agree since $J^2 \in H$.

The charge-one assignment and original Q-grading remain in force. Replacing integer local charge by parity for topological species does not discard the superconducting charge or flux unit, and is not the charge-two construction with an independent electron. Only object, induction and electron-action data are identified here, not the full crossed-braiding or fusion data. Restricting to local A-modules would lose sectors: locality with respect to J^2 only permits charges $0, N/2$, whereas all q are required here. Each orbit and each of its resolved constituents retains the same Q.

Lemma 2 (cyclic resolution count). An H-orbit with stabilizer order h contributes exactly h simple modules.

Proof. The count does not assume that every fixed point splits by its stabilizer in an arbitrary condensation. If $h(a) = |\operatorname{Stab}_H(a)|$, the endomorphism algebra of the induced module is a twisted group algebra of that stabilizer [3, Lemma 4.6 and Proposition 4.7]. The stabilizer is cyclic. Every complex projective representation of a finite cyclic group can be made linear: rescale the operator of a generator so that its appropriate power equals the identity. Its twisted group algebra therefore has $h(a)$ distinct one-dimensional irreducibles. Hence an H-orbit resolves into exactly $h(a)$ distinct simple modules. Different orbits have no common constituent [3]. This answers the untwisted-stabilizer caveat emphasized in [4].

An orbit has $(N/2)/h(a)$ labels. Distributing its $h(a)$ resolved objects among those labels yields the exact formula

$$R_q = \frac{2}{N} \sum_{a \in X_N : Q(a) = q} h(a)^2. \tag{1}$$

Equivalently, a full rotation orbit of length ℓ contributes $2N/\ell$ if ℓ is even, and $N/(2\ell)$ if it is odd. These are the electron-retaining multiplicities of [1]. The proof below does not require determining the electron permutation of each short-orbit resolution label.

3. Exact charge counts

Define $A_L(r)$ as the number of nonnegative L-tuples of sum L whose weighted sum is $r \pmod L$. Let $J_2(d)$ be the second Jordan totient, characterized by $h^2 = \sum_{d|h} J_2(d)$. It is positive: $J_2(d) = d^2 \prod_{p|d}(1 - p^{-2})$.

For $d \mid N/2$, the subgroup of H of order d fixes precisely the labels formed by repeating a block of length $L = N/d$, d times. The block has sum L, and L is even. If its weighted sum is S, the full weighted sum is

$$dS + \frac{L^2 d(d-1)}{2} \equiv dS \pmod N.$$

The second term is divisible by $N = dL$ because L is even. Consequently (1) gives

$$R_q = \frac{2}{N} \sum_{\substack{d | N/2 \\ d | q}} J_2(d) A_{N/d}(q/d). \tag{2}$$

For $q = 0$, every divisor is included.

Lemma 3 (known charge maximum). For $r \neq 0 \pmod L$, $A_L(r) < A_L(0)$. The self-contained derivation below specializes [5, Theorem 1 and Proposition 8] to $n = m = L$, $k = r$. Its strict maximum is also implied by that source's nonnegative divisor decomposition. The finite-group form is [6, Theorem 1] for $\mathbb{Z}/L$ and multiset size L; the neutral count occurs in [7, pp. 38–39].

Proof. Write $c_e(r)$ for the Ramanujan sum over primitive e-th roots raised to the power $-r$. A roots-of-unity filter gives

$$A_L(r) = \frac{1}{L} \sum_{e | L} c_e(r) \binom{2L/e - 1}{L/e}. \tag{3}$$

Indeed, at an L-th root z of exact order e,

$$\prod_{i=0}^{L-1} (1 - t z^i)^{-1} = (1 - t^e)^{-L/e}.$$

Its t^L coefficient is the binomial coefficient in (3). Grouping the root filter by exact order proves the identity.

Every $c_e(r)$ is real and at most $\varphi(e) = c_e(0)$: pair conjugate roots and bound each real part by one. If $r \neq 0 \pmod{L}$, the inequality for $e = L$ is strict, since the primitive root $\exp(2\pi i/L)$ raised to r is not one. All binomial coefficients in (3) are positive. Therefore

$$A_L(r) < A_L(0) \qquad (r \neq 0 \pmod{L}). \tag{4}$$

Compare (2) at $q \neq 0$ and at zero. Each included term is at most the corresponding zero-sector term by (4); omitted terms are nonnegative. The $d = 1$ term is strictly smaller. Thus $R_q < R_0$, uniformly in N and q.

Corollary (absolute rank gap). For every even $N \geq 4$ and $0 < q < N$, $R_0 - R_q \geq N/2$.

Proof. For every $d \mid N/2$ with $d \mid q$, we have $0 < q/d < N/d$. Lemma 3 and integrality imply $A_{N/d}(0) - A_{N/d}(q/d) \geq 1$. If $d \nmid q$, the omitted neutral term satisfies $A_{N/d}(0) \geq 1$, witnessed by the block $(N/d, 0, \ldots, 0)$. Subtracting equation (2) sectorwise therefore gives

$$R_0 - R_q \geq \frac{2}{N} \sum_{d \mid N/2} J_2(d) = \frac{2}{N} \left(\frac{N}{2}\right)^2 = \frac{N}{2}.$$

This is an elementary quantitative consequence of the known strict charge maximum and the resolved-rank formula, not a new counting theory. Equality occurs at $N = 4, q = 2$, since $8 - 6 = 2$. No classification of all equality cases is asserted.

4. Excluding the remaining factor

Lemma 4 (rank–electron obstruction). Suppose the neutral simple objects carry a free physical-electron involution, with $D = R_0/2$ dynamical representatives. Suppose the permitted electron-compatible vortex factors have either two objects exchanged by that electron, or one electron-fixed object, with products counted over the D representatives. A sector of rank different from R_0 containing an electron-moved object admits neither factorization. The conclusion is unchanged by invertible Abelian dressing.

Proof. The two-object product has $2D = R_0$ objects. In the one-object product, the specified electron fixes every product object. Both conclusions contradict the respective hypotheses. Tensoring by an invertible dressing object is a bijection on simple objects and intertwines the specified electron action, so does not alter either obstruction. This criterion is an elementary unpacking of the two allowed factor types, not a classification of other possible factorizations or a new general condensation theorem.

Lemma 5 (full-period electron pair). Every nonzero sector contains distinct simple objects exchanged by the physical electron.

Proof. For any $1 \leq q < N$, take $a_0 = N - 1$, $a_q = 1$, with all other entries zero. This label has charge q. Its unique largest entry forces its full rotation period to be N, so its stabilizer in H is trivial. Its induced module is simple. The rotated label Ta is in a different H-orbit: equality $Ta = T^{2j}a$ would give an odd rotation fixing a full-period label. The two induced modules are distinct and exchanged by the image of J, namely f. This contradicts the one-object factor. Both possible factor types are excluded in every nonzero sector, proving the theorem.

The restriction $N \geq 4$ matters. At $N = 2$, the chosen label is $(1, 1)$, which has no unique largest entry. This is the familiar invertible Ising case and is not part of the claim.

5. Checks, scope and prior work

The boundary case $N = 2$ has ranks $(2, 1)$: it passes the strict rank test but its proposed witness $(1, 1)$ is fixed. It therefore demonstrates why the electron-pair obstruction is indispensable.

Direct label enumeration and the separately implemented divisor formula give full rank vectors $(8, 4, 6, 4)$ for $N = 4$ and $(32, 25, 26, 28, 26, 25)$ for $N = 6$, reproducing [1]. They also agree at $N = 8, 10$; divisor-formula inequalities were checked through $N = 100$. These exact integer calculations detect convention and coding errors, but no finite check substitutes for Sections 2–4. The auxiliary diagnostic counts resolved objects lying over J-invariant H-orbits. It does not determine whether each resolved constituent is individually electron-fixed: electron fusion may permute constituents over an invariant orbit. Neither rank checks nor the full-period witness relies on that diagnostic. Exported rank functions support positive even N, with $N = 2$ retained only as a boundary check; the theorem requires $N \geq 4$.

The family was constructed in [2], and [1] proposes its primitivity while leaving a general proof open. Cyclic module resolution is established machinery [3,4], and the charge count and strict maximum have exact predecessors [5–7]. Our claimed addition is their application with the full-period electron pair, closing that particular all-parameter primitivity gap. We do not claim a new condensation theory, a full classification, computed fusion rules, experimental realization, or an independently assessed contribution rating.

Version control matters for comparison: [2] v1 assigned four objects to the all-ones $N = 4$ fixed point; v4 corrects that count to two, in agreement with [1]. The calculations here retain the physical electron and use the later convention throughout, specifically [2, supplementary equation (S24), PDF p. 11].

Closset and collaborators [8, Section 3.4 and Section 4.1; 9, Section 3.1.4] give close charge, orbit, stabilizer and Jordan-totient counting machinery. Their supersymmetric level is $K = k + N$, so our bosonic level $k = N$ corresponds to $K = 2N$, with gauged subgroup order $N/2$. Their subgroup-neutral quotient counts combine fine charges: the corresponding count is $R_q + R_{q+N/2}$, up to charge sign convention, rather than an individual R_q. For example, the zero-theta counts are $14 = 8 + 6$ at $N = 4$ and $60 = 32 + 28$ at $N = 6$. These printed coarse counts alone do not separate charge zero from $N/2$, or supply the electron-pair obstruction. Their broader formalism remains an important counting predecessor, not a claim of methodological independence here. No complete earlier primitivity proof was found in the bounded source comparison conducted on 11 October 2026; this is not a certification of historical priority.

Declarations

The scholarly creator is Anonymous. AI-assisted preparation includes review remediation, source comparison and internal mathematical checks. No human research contribution is inferred from publication authority. The manuscript is a theoretical candidate, not a claim of microscopic realization, experimental validation or formal verification. Supplied review assessments remain tied to their inspected versions.

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